



Cloud Instance Recommender

Complete User Guide

🔗 AWS EC2

🔗 Azure VMs

🔴 GCP Compute

☁️ Multi-Cloud

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3.0

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👥 IT Professionals · Cloud Architects · DevOps
Engineers





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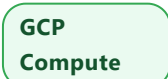
1. Introduction

Welcome to the **Cloud Instance Recommender**, a comprehensive browser-based tool designed to help organisations optimise their cloud infrastructure across AWS, Azure, and Google Cloud Platform. All processing happens entirely in your browser — no data is ever uploaded to an external server.

Version 3.0 builds on the Rule Engine introduced in v2.0 with five new capabilities: **interactive Rule Engine UI dropdowns** that set global defaults without CSV changes, a **Minimum Generation filter** to exclude EOL instance families (e.g., AWS m5/r5), **conflict detection** that highlights contradicting filter combinations in red, a **split AWS Pricing Calculator Bulk Template** that generates separate files for Like-to-Like and Optimized recommendations, and a **Min Gen CSV column** for per-row generation control.

Benefit	Description
 Right-sizing	Identify over-provisioned VMs and recommend appropriately-sized replacements
 Smart Scaling	N/2, N, N+1 strategy with industry-standard 40 % / 80 % thresholds
 Rule-based guardrails	Production environments automatically block burstable, prev-gen, and undersized instances
 OS Compatibility	Windows workloads automatically exclude ARM/Graviton instances that cannot run them
 Multi-Cloud	Side-by-side recommendations across AWS, Azure, and GCP in one run
 Audit trail	"Rules Applied" output column explains every recommendation decision
 Privacy-first	100 % client-side — your inventory data never leaves your browser

Supported Cloud Providers

Provider	Regions	Instance Families	Notable Support
 AWS EC2	35 regions	t, m, c, r, x, z, p, g, i, d, trn, inf, mac, hpc...	Graviton (ARM), Nitro Enclaves, Bulk Template export
 Azure VMs	63 regions	B, D, E, F, H, L, M, N, NC, ND, NV...	ARM (DpdsV5, Bpsv2), Burstable B-series detection
 GCP Compute	47 regions	E2, N2, N4, C2, C3, C4, M1–M4, A2, A3, G2, T2A, H3, Z3...	T2A ARM, shared-core detection, zone normalisation



2. Getting Started



System Requirements

Requirement	Specification
Browser	Chrome 80+, Firefox 75+, Edge 80+, Safari 13+
JavaScript	Must be enabled
CSV file size	Up to 10 MB ($\approx$ 5,000–10,000 VMs)
Internet	Required for initial page load; all processing is local thereafter
Local storage	Used for usage statistics only; no sensitive data stored



Accessing the Platform

The tool is hosted on GitHub Pages and requires no installation:

- 1 Open your browser and navigate to the Cloud Instance Recommender URL.
- 2 From the homepage, select your target: **AWS** , **Azure** , **GCP** , or **Multi-Cloud** .
- 3 Each provider page loads independently with provider-specific filtering options.

TIP

The data files are large (AWS ~5 MB, Azure ~10 MB, GCP ~3 MB). On first load, allow a few seconds for the browser to parse the instance catalogue before uploading your CSV.

3. Data Preparation

Required Columns

Every CSV row must contain the following columns. Column names are auto-detected (case-insensitive, spaces/underscores normalised).

Column Name	Description	Example
VM Name	Unique identifier for the virtual machine	web-server-01
CPU Count	Number of vCPUs currently allocated	4
Memory (GB)	RAM in gigabytes currently allocated	16
AWS Region / Azure Region / GCP Region	Target cloud region for recommendation (provider-specific column)	us-east-1 / East US / us-central1-a
CPU Utilization	Average CPU utilisation % — required for Optimized recommendations	45
Memory Utilization	Average memory utilisation % — required for Optimized recommendations	60

Optional Columns

These five columns activate the **Rule Engine**. All are optional — blank cells use the safe defaults shown below. They can also be set globally via the Rule Engine UI dropdowns in the Advanced Filtering section (CSV per-row values always override the UI defaults).

Column	Accepted Values	Default	What it activates
ENV	Production , Prod , Staging , Stage , Dev , Test	No rules	Rules 1a, 1b, 1c, 1d — see Section 4
OS	Linux , Windows , Windows Server , macOS	Linux	ARM/Graviton exclusion for Windows; mac1/mac2-only for macOS (AWS)
Workload	General , Database , Web Server , Cache , ML/AI , Batch , HPC	General	Preferred instance families sorted first before cheapest selection
Compliance	PCI , HIPAA , FIPS	None	Forces current-gen + Nitro Enclaves (AWS PCI/HIPAA)
Min Gen	AWS 5 , 6 , 7 Azure 3 , 4 , 5 (v-number) GCP n2 , n2d , n4	No minimum	Excludes instances older than the specified generation (e.g., AWS m5 is excluded when Min Gen = 6)

HOW RULES ARE APPLIED

Rules are evaluated per-row — each VM in your CSV can have different ENV, OS, Workload, and Compliance values. A

RULES APPLIED

column in the output CSV documents exactly which rules fired for each VM.

Sample CSV Structure

```
VM Name,CPU Count,Memory (GB),CPU Utilization,Memory Utilization,AWS Region,ENV,OS,Workload,Compl
web-server-01,4,16,45,60,us-east-1,Production,Linux,Web Server,,6
db-server-02,8,32,70,80,us-west-2,Production,Windows,Database,PCI,7
app-server-03,2,8,35,45,eu-west-1,Dev,Linux,General,,
cache-server-04,4,16,25,30,us-east-1,Staging,Linux,Cache,,6
ml-server-05,8,64,80,75,us-west-2,Production,Linux,ML/AI,HIPAA,7
```

FORMATTING RULES

Save files as UTF-8 CSV. Do not include currency symbols or % signs in numeric columns — the tool strips them automatically. Region names must match the exact format shown in the in-app sample template.

Data Collection Tips

Provider	Where to get utilisation data
AWS	CloudWatch → EC2 → CPUUtilization metric (14–90 day average recommended). Cost Explorer for current instance types.
Azure	Azure Monitor → Virtual Machines → Percentage CPU & Available Memory. Azure Advisor for existing recommendations.
GCP	Cloud Monitoring → compute.googleapis.com/instance/cpu/utilization . Recommender API for existing rightsizing signals.

Collection best practices:

- ✓ Use a 30–90 day average — short windows miss weekly/monthly patterns
- ✓ Include at least one peak business period (quarter end, product launch, etc.)
- ✓ Use P95 or P99 if your workload has spiky traffic, not just averages
- ✓ Validate CPU count and memory against the actual current instance type, not OS-reported values

🛡️ 4. Environment & Workload Rules

The Rule Engine applies automatically when optional CSV columns are present. Rules narrow the candidate instance list *before* the cheapest-match selection runs — so every recommendation already satisfies your environment constraints.

⚡ Rule 1a — Burstable Exclusion

When: ENV = Production or Staging

Effect: Removes CPU-credit-based families

AWS t1, t2, t3, t3a, t4g

Azure B-series (bsv2, bsv3, bpsv2...)

GCP f1-micro, g1-small, e2 shared-core

🕒 Rule 1b — Generation & Compliance

When: ENV = Production OR Compliance set

Effect: Current-generation instances only

Additionally for **AWS** PCI/HIPAA:

Nitro Enclaves support required

Rationale: Prev-gen lacks modern security features

🏠 Rule 1c — Minimum Size Floor

When: ENV = Production or Staging

Effect: Excludes undersized instances

AWS Excludes nano and micro sizes

Azure ≥ 2 vCPUs (Production)

GCP ≥ 2 vCPUs (Production)

Rationale: Headroom for monitoring agents and traffic spikes

🌐 Rule 1d — Network Preference

When: ENV = Production AND Workload = Database or Web Server

Effect: Prefers instances with ≥ 4 vCPUs

Rationale: Larger instance sizes map to higher network bandwidth tiers, critical for database replication and high-throughput web traffic

📁 OS — Compatibility Rules

OS = Windows / Windows Server:

Excludes all ARM/Graviton instances (t4g, m6g, c6g, r6g on AWS; Dpdsv5, Bpsv2 on Azure; T2A on GCP)

OS = macOS **AWS only**:

Limits to mac1 and mac2 families

🎯 Workload — Family Preference

Sorts recommended families to the top before cheapest-selection runs. Falls back to cheapest overall if no preferred family has matching instances.

1 2 3 4 Min Generation Rule (MG)

The Min Generation rule excludes instances older than a specified hardware generation. This is useful when older generations are approaching end-of-life (e.g., AWS m5/r5 EOL) or when you have a policy requiring modern silicon.

Provider	Column / UI value	Excluded when set to...	Included families
 AWS	5 , 6 , 7	Min Gen = 6 → excludes m5, r5, c5, t3, and all earlier gens	m6i, m6a, m7i, m7g, r6i, r7a, c6i, c7g...
 Azure	3 , 4 , 5 (v-number)	Min Gen = 4 → excludes Dsv3, Esv3 and older	Dsv4, Dsv5, Esv4, Esv5, Fsv2...
 GCP	n2 , n2d , n4 (family name)	Min Gen = n2 → excludes N1, E2 shared-core, F1, G1	N2, N2D, C2, C2D, T2A, A2, G2, C3, N4, C4...

MULTI-CLOUD MIN GEN

On the Multi-Cloud page, Min Gen uses AWS generation numbers (5/6/7) as a unified reference. Azure values are mapped by subtracting 2 (AWS 6 → Azure v4), and GCP values are mapped to the corresponding generation tier.

Workload Family Mappings

Workload	AWS Preferred Families	Azure Preferred Series	GCP Preferred Series
General	m (General Purpose)	D-series	N2, E2
Database	r, x, z (Memory-optimised)	E-series, M-series	M1, M2, M3, M4
Web Server	m, c (General + Compute)	D-series, F-series	N2, E2, N4
Cache	r, x (Memory-optimised)	E-series, M-series	M1, M2, M3
ML/AI	p, g, trn, inf (GPU/Accelerated)	NC, ND, NV series	A2, A3, G2
Batch	c, m (Compute-optimised)	F-series, D-series	C2, C2D, C3, C3D
HPC	hpc, c	HB, HC series	H3, C2

Rules Reference Summary

Rule	Trigger	Action	Output tag
1a	ENV = Production or Staging	Remove burstable families	1a: Burstable excluded
1b	ENV = Production or Compliance set	Current-gen only	1b: Prev-gen excluded
1b-Nitro	Compliance = PCI/HIPAA + AWS	Nitro Enclaves required	1b: Nitro required (Compliance)
1c	ENV = Production or Staging	Size floor (no nano/micro)	1c: Size floor applied
1d	ENV = Production + DB/Web Workload	≥ 4 vCPUs preferred	1d: Network-tier preference (≥4 vCPUs)
OS-Win	OS = Windows	Exclude ARM/Graviton	OS: ARM excluded (Windows)
OS-Mac	OS = macOS (AWS only)	mac1/mac2 families only	OS: mac1/mac2 only (macOS)
Workload	Workload ≠ General	Preferred families sorted first	Workload: database preference
MG	Min Gen set (CSV or UI)	Exclude instances below specified generation	MinGen: 6+

FALLBACK BEHAVIOUR

If a combination of rules eliminates all candidate instances (e.g., Production + PCI + very small CPU/memory requirement), the engine notes this in the Rules Applied column with a warning and falls back to the closest available instance. Check the Rules Applied column if you see unexpected recommendations.



5. Step-by-Step Usage Guide

Step 1 — Choose Your Cloud Provider

- **Single provider:** Navigate directly to the AWS, Azure, or GCP page for focused recommendations with provider-specific filtering options.
- **Multi-cloud:** Use the Multi-Cloud page to compare recommendations across all three providers in a single run from one CSV upload.

Step 2 — Download and Prepare the Template

- 1 Click **Download Sample CSV** on the provider page to get a template with the correct column headers.
- 2 Replace the sample rows with your actual VM inventory. Include the optional ENV, OS, Workload, and Compliance columns where applicable.
- 3 Save the file as UTF-8 CSV.

Step 3 — Upload Your CSV

- 1 Drag and drop your CSV onto the upload zone, or click it to browse.
- 2 The tool automatically validates the file, checks required columns, shows a data preview, and assigns a quality score (0–100 %).
- 3 Review any warnings before proceeding — common issues include missing region columns or non-numeric CPU/memory values.

Quality Score	Level	Meaning
90–100 %	 Excellent	Complete data, minimal issues
70–89 %	 Good	Some missing optional values
50–69 %	 Fair	Multiple data quality issues
< 50 %	 Poor	Significant missing or invalid data

Step 4 — Select Recommendation Type

Type	What it does	When to use	Data needed
Like-to-Like	Finds cheapest instance with $\geq$ current CPU and $\geq$ current memory	Conservative migrations; guaranteed headroom	CPU Count, Memory (GB), Region
Optimized	Applies N/2, N, N+1 strategy to right-size based on actual utilisation	Cost reduction; over-provisioned estate	Above + CPU Utilization, Memory Utilization
Both	Generates both columns per VM for comparison	Decision-making; stakeholder reporting	All columns

Step 5 — Configure Optimization Settings

(Only relevant for Optimized or Both recommendation types)

Optimization Mode

- **CPU-Based:** Scale CPU according to utilisation thresholds
- **Memory-Based:** Scale memory according to utilisation thresholds
- Both can be enabled simultaneously

N/2, N, N+1 Thresholds (Industry-Standard Defaults)

Zone	CPU / Memory Utilisation	Action	Rationale
Downsize (N÷2)	$\leq 40\%$	Halve the target resource	Avg below 40 % = significantly over-provisioned; AWS, Azure, and GCP advisors all flag this range
Keep Same (N)	40 % – 80 %	Use current resource count	Right-sized zone with adequate headroom for spikes
Upsize (N+1)	$> 80\%$	Add one unit of the resource	SRE best practice: keep utilisation below 80 % to avoid saturation incidents

THESE DEFAULTS ARE EDITABLE

The downsize and keep-max thresholds can be adjusted in the UI. For example, if you use P95 utilisation data (not averages), you may want to lower the downsize threshold to 30 %.

Step 6 — Configure Rule Engine & Advanced Filtering (Optional)

Expand the **Advanced Filtering** section to access both the Rule Engine UI controls and provider-specific filters.

Rule Engine Defaults

Five dropdowns let you set rule defaults for the entire batch without modifying your CSV. Per-row CSV column values always take priority over these UI defaults.

Dropdown	Options	Effect
Default Environment	— / Production / Staging / Dev / Test	Applies ENV rules to every row that has no ENV column value
Default OS	— / Linux / Windows / macOS	Applies OS compatibility rules globally
Default Workload	— / General / Database / Web Server / Cache / ML/AI / Batch / HPC	Sets family preference order for all rows
Default Compliance	— / PCI / HIPAA / FIPS / SOC2	Enforces compliance-grade filtering globally
Minimum Generation	AWS: Gen 5+ / 6+ / 7+ · Azure: v3+ / v4+ / v5+ · GCP: N2 / N2D / N4	Excludes instance families below the selected generation

Conflict Detection

When two selected options directly contradict each other, the conflicting dropdowns are highlighted in red with an explanatory warning. For example:

- OS = Windows + processor filter restricted to Graviton/ARM only
- ENV = Production + main families restricted to burstable types only (t/B/e2)
- Workload = ML/AI + processor filter restricted to Intel/AMD only (no GPU available)

Resolve conflicts before generating recommendations — rules will still run but may produce unexpected results if contradictions are left in place.

Step 7 — Generate and Download Results

- 1 Click  **Generate Recommendations**. A progress bar shows processing status for large files.
- 2 Once complete, the download section updates based on what was generated.
- 3 **AWS page** — **both Like-to-Like and Optimized selected:** three buttons appear:  **Download Results CSV** — all columns in one file  **Bulk Template (Like-to-Like)** — AWS Pricing Calculator format, Like-to-Like instances only  **Bulk Template (Optimized)** — AWS Pricing Calculator format, Optimized instances only
- 4 **AWS page — single type selected:** Results CSV + one Bulk Template button.
- 5 **Azure / GCP pages:** Results CSV only.

6. Understanding Recommendations

Like-to-Like Strategy

- 1 The full instance catalogue for the specified region is loaded.
- 2 All instances with vCPUs < required CPU or memory < required memory are removed.
- 3 The Rule Engine applies ENV, OS, Workload, and Compliance filters.
- 4 The cheapest remaining instance is returned (workload preferred families are sorted to the top first).

N/2, N, N+1 Optimization Strategy

- 1 Calculate target CPU and memory from utilisation data using the 40 %/80 % thresholds.
- 2 Run the same Like-to-Like process against the *adjusted* targets.

Scenario	CPU Util	Memory Util	Target CPU	Target Memory
Over-provisioned	25 %	30 %	Current ÷ 2	Current ÷ 2
Right-sized	55 %	65 %	Current (N)	Current (N)
Under-provisioned	85 %	40 %	Current + 1	Current (N)
Mixed	25 %	85 %	Current ÷ 2	Current + 1

7. Output Columns Explained

The downloaded CSV preserves all original input columns and appends recommendation columns for each selected provider.

Original Input Columns (preserved)

VM Name, CPU Count, Memory (GB), CPU Utilization, Memory Utilization, [Provider] Region, ENV, OS, Workload, Compliance — all returned unchanged.

Recommendation Columns (added per provider)

Column	Description	Example value
AWS Like-to-Like Instance	Recommended EC2 instance type (like-for-like)	m6i.xlarge
AWS Like-to-Like vCPUs	vCPU count of the recommended instance	4
AWS Like-to-Like Memory (GiB)	Memory of the recommended instance in GiB	16
AWS Optimized Instance	Recommended EC2 instance type (optimised)	m6i.large
AWS Optimized vCPUs	vCPU count of the optimised instance	2
AWS Optimized Memory (GiB)	Memory of the optimised instance in GiB	8
AWS Rules Applied	Pipe-separated list of rules that fired for this row	1a: Burstable excluded 1c: Size floor applied

The same pattern applies for Azure and GCP (replacing "AWS" with the provider name).

WHY NO PRICING IN THE OUTPUT?

Cloud pricing changes frequently and varies with discounts, reserved instances, Savings Plans, and enterprise agreements. This tool uses pricing internally to rank candidates and find the cheapest match, but does not surface prices in the output CSV to avoid giving users stale or context-free figures. Always verify pricing in the official AWS/Azure/GCP pricing calculators before making commitments.

Status values for recommendation columns

Value	Meaning
m6i.xlarge (etc.)	Successful recommendation
Missing data	Required input column (CPU, Memory, or Region) was blank
No utilization data	Optimized type selected but CPU/Memory Utilization were 0 or blank
No data available	No instance in the region catalogue met the requirements (after rules applied)
Error	Unexpected processing error — check browser console for details



8. Export Options



Standard Results CSV

Available on all provider pages. Downloads a CSV containing all original input columns plus the recommendation columns described in Section 7. This file is ready for:

- ✓ Excel / Google Sheets pivot table analysis
- ✓ Importing into ITSM or CMDB tools
- ✓ Sharing with stakeholders for review
- ✓ Filtering by "Rules Applied" to audit environment-specific decisions



AWS Pricing Calculator Bulk Template

AWS only

Available exclusively on the **AWS page** after recommendations are generated. Produces a CSV that exactly matches the column schema of the *Amazon EC2 Instances* worksheet in the AWS Pricing Calculator's Bulk Upload Template.

Column mapping

Bulk Template Column	Populated from	Default
Group	ENV column value	Default
Description	VM Name	—
AWS Region	AWS Region column	—
Operating System	OS column (Windows → "Windows Server", else "Linux")	Linux
Instance Type	Like-to-Like or Optimized instance type (one per file)	—
Tenancy	Fixed	Shared Instances
Number of Instances	Fixed	1
Usage Type	Fixed	Always On
Purchasing Options	Fixed	On-Demand
Storage columns	Left blank for manual completion	—

ONE FILE PER RECOMMENDATION TYPE

When both Like-to-Like and Optimized are generated, two separate bulk template buttons appear — one for each type. Always import only one file per estimate so you don't double-count instances and inflate the pricing figure.

How to import into AWS Pricing Calculator

- 1 Open the [AWS Pricing Calculator](#) .
- 2 Create a new estimate or open an existing one.
- 3 Click **Add service** → **Amazon EC2** .
- 4 Choose **Add group** → **Bulk import** .
- 5 Upload the CSV file downloaded from this tool (Like-to-Like *or* Optimized — not both).
- 6 Review the pre-populated rows, add storage details as needed, and save the estimate.

9. Advanced Filtering

The Advanced Filtering section contains two types of controls: the **Rule Engine UI** (global defaults for ENV/OS/Workload/Compliance/Min Gen) and **provider-specific filters** that narrow the instance candidate pool. Provider filters are applied first; the Rule Engine then runs on the already-filtered set.

Rule Engine UI Controls

Five dropdowns inside Advanced Filtering set rule defaults for the entire batch. These act as a fallback when a CSV row has no value in the corresponding column — per-row CSV values always override them.

Dropdown	AWS options	Azure options	GCP options
Default Environment	Production / Staging / Dev / Test (same across all providers)		
Default OS	Linux / Windows / macOS (same across all providers)		
Default Workload	General / Database / Web Server / Cache / ML/AI / Batch / HPC		
Default Compliance	PCI / HIPAA / FIPS / SOC2		
Minimum Generation	Gen 5+ / Gen 6+ / Gen 7+	v3+ / v4+ / v5+	N2 / N2D / N4

Conflict Detection

When a UI selection directly contradicts another filter, the conflicting control is highlighted with a red border and an inline warning message. Three conflict types are detected:

Conflict	What triggers it	Highlighted control
OS ↔ Processor	OS = Windows AND all selected processor filters are ARM/Graviton-only	Default OS dropdown
ENV/Compliance ↔ Families	ENV = Production or Compliance set, AND main family filter includes only burstable types (t/B/e2)	Default Environment or Default Compliance dropdown
Workload ↔ Processor	Workload = ML/AI AND processor filter excludes all GPU/ARM options	Default Workload dropdown

CONFLICTS ARE WARNINGS, NOT BLOCKERS

The tool will still generate recommendations when conflicts exist, but the results may not match your intent. Resolve the conflict before generating for best results.

AWS-Specific Filters

Filter	Description
Current Generation Only	Excludes previous-generation families (t2, m4, c4, r4, etc.). Recommended for all new deployments.
Instance Family Names	Restrict to specific family categories: General Purpose, Compute Optimised, Memory Optimised, Storage Optimised, Accelerated Computing
Processor Manufacturer	Intel, AMD, or AWS (Graviton/ARM). Note: selecting AWS Graviton and then setting OS = Windows via the CSV column will still exclude Graviton — OS rules always take priority.
Main Family Prefix	Restrict by family prefix (t, m, c, r, x, z, p, g, i, d...)
Exclude Types	Exclude individual type classes: Graviton/ARM, GPU, Mac instances, previous generation, Nitro-only

◆ Azure-Specific Filters

Filter	Description
VM Series	Filter by series: B (burstable), D (general), E (memory), F (compute), H (HPC), L (storage), M (large memory), N (GPU)
Processor Architecture	Intel x86-64, AMD EPYC, ARM (Ampere)
VM Families	Filter by specific family strings (dv5, ev5, fsv2...)
Exclude Types	Exclude ARM, GPU, or burstable categories

● GCP-Specific Filters

Filter	Description
Machine Families	E2, N1, N2, N4, C2, C2D, C3, C3D, C4, M1, M2, M3, M4, A2, A3, G2, T2D, T2A, H3, Z3
CPU Platform	Intel, AMD, ARM
Machine Type Prefix	Filter by standard, highmem, highcpu, ultramem
Exclude Types	Exclude ARM (T2A), GPU, or shared-core families

10. Multi-Cloud Comparison

The Multi-Cloud page processes a single CSV against all three providers simultaneously, producing side-by-side recommendations in one output file. This is useful for migration planning, vendor selection, and lock-in risk assessment.

CSV Format for Multi-Cloud

Include region columns for every provider you want to compare. Providers with a blank or missing region column are skipped for that row.

```
VM Name,CPU Count,Memory (GB),CPU Utilization,Memory Utilization,AWS Region,Azure Region,GCP Region
web-server-01,4,16,45,60,us-east-1,East US,us-central1-a,Production,Linux,Web Server
db-server-02,8,32,70,80,us-west-2,West US 2,us-west1-b,Production,Linux,Database
```

Regional Equivalents Reference

Geography	AWS Region	Azure Region	GCP Region
US East	us-east-1	East US	us-east1-b
US West	us-west-2	West US 2	us-west1-b
Europe West	eu-west-1	North Europe	europa-west1-c
Europe Central	eu-central-1	Germany West Central	europa-west3-a
Asia Pacific (SE)	ap-southeast-1	Southeast Asia	asia-southeast1-a
Asia Pacific (NE)	ap-northeast-1	Japan East	asia-northeast1-a

Interpreting Multi-Cloud Output

The output CSV contains recommendation columns for every selected provider in the same row, making it easy to filter in Excel or Google Sheets:

- ✓ Sort by VM Name to see all providers for each workload
- ✓ Use the "Rules Applied" columns to understand why providers differ
- ✓ Provider pricing calculators should be used to compare actual costs — do not compare instance types directly as the vCPU/memory ratio differs across providers

11. Troubleshooting

File Upload Problems

Problem	Likely Cause	Solution
"CSV parsing failed"	Special characters, wrong encoding, or malformed CSV	Save the file as UTF-8 CSV (not UTF-8 BOM). Remove characters like <code><</code> <code>></code> <code>&</code> from data values. Check for unescaped commas inside quoted fields.
"File too large"	CSV exceeds 10 MB	Split into smaller batches. Remove unused columns. Filter to the VMs you need recommendations for.
"Missing required columns"	Column name mismatch	Column names are matched case-insensitively and with space/underscore normalisation, but the exact words must be present. Download the sample CSV template and compare your headers.
"Invalid region name"	Typo or wrong provider format	AWS regions use hyphens: <code>us-east-1</code> . Azure uses display names: <code>East US</code> . GCP uses zone format: <code>us-central1-a</code> .

Recommendation Problems

Problem	Cause	Solution
"No data available" for all VMs	Region not supported or data file not yet loaded	Wait for the data files to finish loading (check the browser tab title). Verify the region name is in the supported list (see in-app sample template).
Recommendations seem too large	Utilisation data is missing or zero — falling back to Like-to-Like	Ensure CPU Utilization and Memory Utilization columns contain non-zero values when using Optimized mode.
No recommendation after ENV = Production	Rule engine eliminated all candidates	Check the Rules Applied column. Try relaxing the Advanced Filtering options. Ensure the region has current-generation instances (all major regions do).
Windows VMs getting ARM recommendations	OS column is blank or misspelled	Set the OS column to exactly <code>Windows</code> or <code>Windows Server</code> for affected rows.
Optimized same as Like-to-Like	Utilisation in the 40–80 % "keep same" zone	This is expected — the N/2,N,N+1 strategy keeps the same size when utilisation is in the right-sized band.

Browser / Performance Problems

Problem	Solution
Page feels slow on first load	The data files are large (AWS ~5 MB, Azure ~10 MB). GitHub Pages serves them with gzip compression, but parsing still takes a few seconds. Wait for the browser to finish before uploading.
Download button does nothing	Check that your browser's pop-up blocker is not intercepting the download. Allow pop-ups for the site or use a different browser.
Processing hangs on large files	Split files larger than 2,000 rows into batches. Close other browser tabs to free memory. Refresh the page (this also clears the loaded instance data — allow time to reload).
Results CSV opens as garbled text in Excel	When opening a CSV in Excel, use Data → From Text/CSV and select UTF-8 encoding. Do not double-click the file directly.

🌟 12. Best Practices

Data Collection

- ✓ Collect at least 30 days of utilisation data; 90 days is ideal to capture seasonal variation
- ✓ Use the P95 or P99 percentile for CPU if your workload is spiky — average metrics underestimate peak demand
- ✓ Memory utilisation from the OS is more reliable than hypervisor-reported figures — use OS-level agents where possible
- ✓ Tag VMs with their ENV, OS, and Workload in your CMDB and export that metadata directly into the CSV

Recommendation Implementation

- ✓ Start with Dev/Test environments — fast to change and low risk if something is wrong
- ✓ For Production, use Like-to-Like first to establish a safe baseline, then consider Optimized for a second wave
- ✓ Always have a rollback plan: snapshot or image the VM before resizing
- ✓ Monitor for 48–72 hours after a resize before treating it as complete

Cost Strategy

- ✓ Use this tool to identify the right instance *type* — then separately apply Reserved Instance or Savings Plan pricing in the provider's pricing calculator
- ✓ Re-run the tool quarterly for dynamic workloads; semi-annually for stable production systems
- ✓ After a cloud migration, re-run with actual utilisation data from the new environment — the right-sizing picture often changes after migration

Security & Compliance

- ✓ Set `Compliance = PCI` or `Compliance = HIPAA` in the CSV for regulated workloads to ensure Nitro-capable (AWS) and current-gen-only instances are recommended
- ✓ For FIPS-boundary requirements, manually verify that recommended regions are within your geographic compliance boundary
- ✓ No inventory data ever leaves your browser — the tool is fully client-side

? 13. Frequently Asked Questions

☀ General

Q: Is my data secure?

A: Yes. All processing happens in your browser — your VM inventory never leaves your machine. The tool loads a static instance catalogue at start-up; there is no backend server receiving your data.

Q: Why doesn't the output include pricing?

A: Cloud pricing is not static — it changes with region, OS, discounts, Savings Plans, Reserved Instances, and enterprise agreements. Showing a price in the output would be misleading within weeks. This tool identifies the right *instance type*; use your provider's pricing calculator for accurate cost figures. The AWS Pricing Calculator Bulk Template export makes this workflow seamless for AWS.

Q: Can I use this tool offline?

A: The initial page load requires an internet connection to download the instance data files. Once loaded, you can process CSV files without connectivity. The instance data is cached in your browser until you refresh the page.

Q: Which file formats are supported?

A: CSV (comma-separated values) with UTF-8 encoding. Excel files must be saved as CSV first.

⚙ Technical

Q: Why does the tool recommend a different instance than our cloud provider's own advisor?

A: Provider advisors (AWS Trusted Advisor, Azure Advisor) typically look at longer time horizons (14–90 days) and use conservative thresholds. This tool applies your chosen utilisation window and thresholds. The recommendations can legitimately differ — treat both as data points rather than absolute answers.

Q: Why are the downsize thresholds set to 40 % instead of 50 %?

A: The 40 %/80 % thresholds align with the approach used by AWS Cost Optimisation, Azure Advisor, and GCP Recommender. At 40 %, there is still comfortable headroom for traffic spikes and OS/monitoring overhead. The old 50 % default was too aggressive for most production workloads. You can adjust both thresholds in the UI if your environment requires different values.

Q: I set ENV = Production but got a t3.large recommendation. Why?

A: Check the Rules Applied column — the rule engine reports exactly what happened. If "1a: Burstable excluded" is not listed, verify that the ENV column value is exactly **Production** (no extra spaces or different capitalisation).

Q: Can I use a mix of ENVs in the same CSV?

A: Yes — rules are applied per row. You can have Production, Staging, and Dev rows in the same file; each gets its own rule evaluation.

Q: What happens if no instance survives after all rules are applied?

A: The tool notes the failure in the Rules Applied column with a ⚠ warning and falls back to the best available instance from the pre-rules candidate set. This is a signal to either relax filters or review whether the requirements are achievable in that region.

 Business

Q: What typical savings can I expect?

A: Organisations commonly see 15–40 % reduction in compute spend from right-sizing alone, before applying Reserved Instances or Savings Plans. The actual figure depends heavily on how over-provisioned the original fleet is.

Q: How often should I re-run the tool?

A: Monthly for dynamic/seasonal workloads; quarterly for stable production workloads; immediately after any significant architectural change or traffic pattern shift.

Q: Can I share results with stakeholders who don't have access to the tool?

A: Yes — the downloaded CSV can be opened in Excel or Google Sheets. The "Rules Applied" column provides plain-English audit notes explaining every recommendation, making it straightforward to present to non-technical stakeholders.

Q: Is the AWS Pricing Calculator Bulk Template ready to import immediately?

A: It populates all required columns with On-Demand, Always On, Shared Instances defaults. You will need to manually add storage details (EBS volume type, size, IOPS) in the calculator after import if required for your estimate.

14. Conclusion

The Cloud Instance Recommender gives you a fast, privacy-first starting point for cloud rightsizing at any scale — from a handful of VMs to thousands. By combining flexible like-to-like and optimised recommendation modes with a configurable rule engine, it produces recommendations that are not just cheap, but appropriate for the environment, OS, workload type, and compliance posture of each individual VM.

Key things to remember:

- ✓ Add ENV, OS, Workload, and Compliance columns to get environment-aware recommendations with a full audit trail
- ✓ Use the 40 %/80 % thresholds as a starting point — adjust them based on whether your utilisation data is averages or percentiles
- ✓ Pricing is intentionally excluded from output — use the AWS Pricing Calculator Bulk Template export for cost estimation
- ✓ Treat recommendations as an informed starting point, not a final answer — validate in a non-production environment first
- ✓ Re-run quarterly to keep pace with changing workloads and the cloud providers' expanding instance catalogues

HAPPY OPTIMISING!

Cloud right-sizing is an ongoing discipline, not a one-time exercise. Build it into your regular operational cadence and it will continue to compound savings over time.